



Urgency Messaging Strategy Analysis

How hotel prices move as check-in approaches — and what business opportunities arise across 5 cities

Saarang Ahuja | Supply Analytics Case Study

49,061

Total Bookings

5

Cities Analyzed

880

Unique Properties

13,236

Chain Hotel Bookings

35,825

Non-Chain Bookings

**Oct–Dec
2016**

Check-in Date Range

Dataset Limitations & Key Assumptions

Dataset Limitations

- Three months of data (Oct–Dec 2016) — seasonal effects may not reflect full-year behaviour.
- No competitor pricing data — Agoda’s price positioning relative to the market is unknown.
- No booking funnel data — conversion rates at each stage cannot be directly measured.
- No occupancy/availability data — scarcity messaging cannot be validated from this dataset.
- City IDs only — market context (resort vs urban vs business) is inferred, not labelled.
- Price direction classification limited to 565 of 880 properties — 315 lacked sufficient bookings in both early and last-minute windows. These contributed to all other analyses.

Key Assumptions & Methodology

Lead Time Formula

Lead Time (days) = Check-in Date – Booking Date. 3 records with negative values excluded.

Lead Time Buckets — Rationale

Buckets reflect natural planning horizons: same-day, last-minute (1–3d), short-advance (4–7d), medium (8–14d), planned (15–30d, 31–60d). Interpretability was prioritized over k-means.

Why Median ADR, Not Mean

Median ignores extreme outliers and shows what the typical booker paid — more honest than mean which is pulled by luxury outliers.

Key Findings

-22.7%

Average Drop in ADR (31–60 Days Ahead → Same-Day)

Average fall in ADR from the 31–60 day window to same-day booking, across all 5 cities:
City A: -12.9% | City B: -30.9% | City C: -50.6% | City D: -6.5% | City E: -12.8%

34%

High-Volume Last-Minute Window

34% of all bookings across 5 cities are made within 3 days of check-in. City E peaks at 44%; City C is lowest at 18% — each city needs a calibrated urgency strategy.

456 of 880

Properties Confirmed to Price Last-Minute Lower or Flat

456 unique properties (52% of all 880) confirmed to price the same or lower last-minute. 315 lacked sufficient data to classify. Only 109 genuinely raise prices last-minute.

CITY-BY-CITY SNAPSHOT

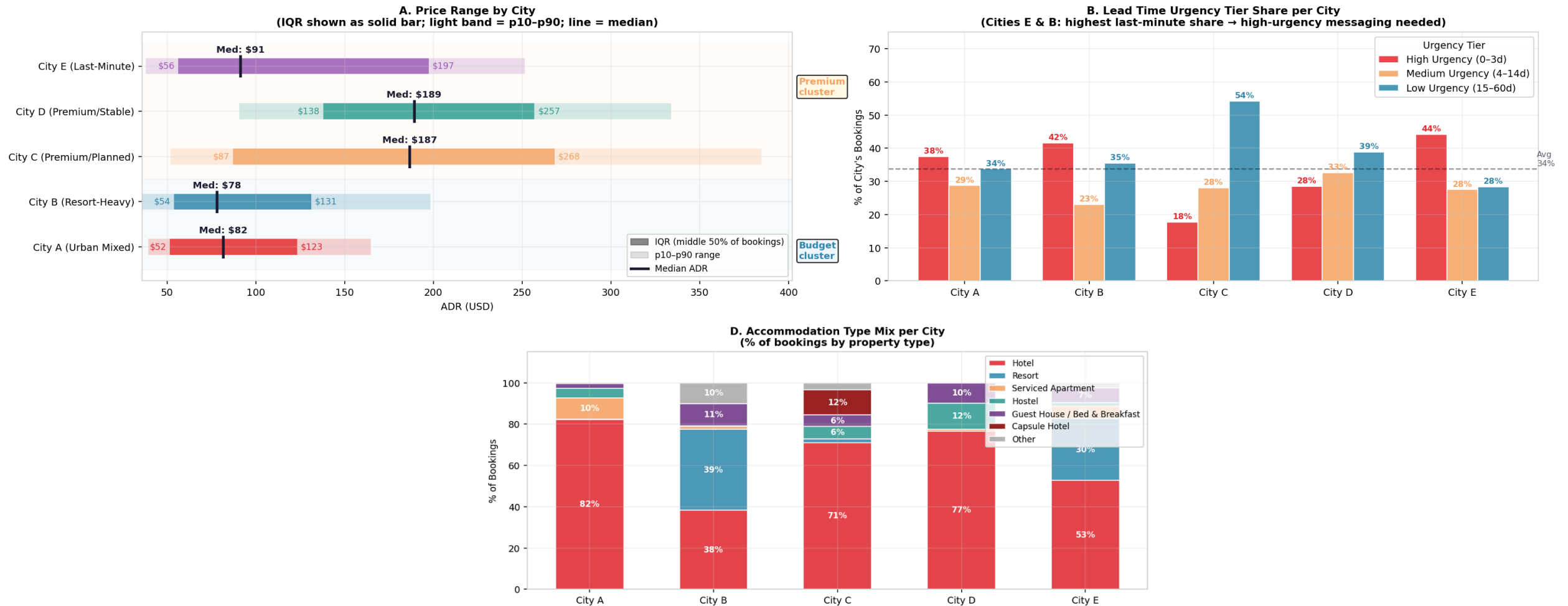
City	ADR Drop (31–60d→same-day)	3-day booking share	% Rising	Market
City A	-12.9%	37.5%	13%	Urban mixed
City B	-30.9%	41.6%	20%	Resort-heavy
City C	-50.6%	17.8%	18%	Premium, plans ahead
City D	-6.5%	28.5%	31%	Premium, price-stable
City E	-12.8%	44.1%	24%	Most last-minute

⚠ Data note: 315 of 880 properties lacked sufficient bookings in both the early (15+ days) and last-minute (0–3 days) windows to classify price direction. These properties still contributed to all other analyses (ADR trends, EDA, segmentation)

Recommendations

- Avoid blanket “prices rising” urgency claims — only truthful for 109 of 880 properties; the majority price last-minute lower or flat
- Use specific types of urgency messaging for different scenarios — for example, use “earn more loyalty points” urgency messaging for budget and mid-range properties as their prices fall further; honest framing captures the booking, builds goodwill and earns customer loyalty
- 4★ properties are the highest revenue segment in 4 of 5 cities (A, B, D, E); in City C where 3★ dominates, the same logic applies — A/B testing should validate the optimal message type per city before full rollout

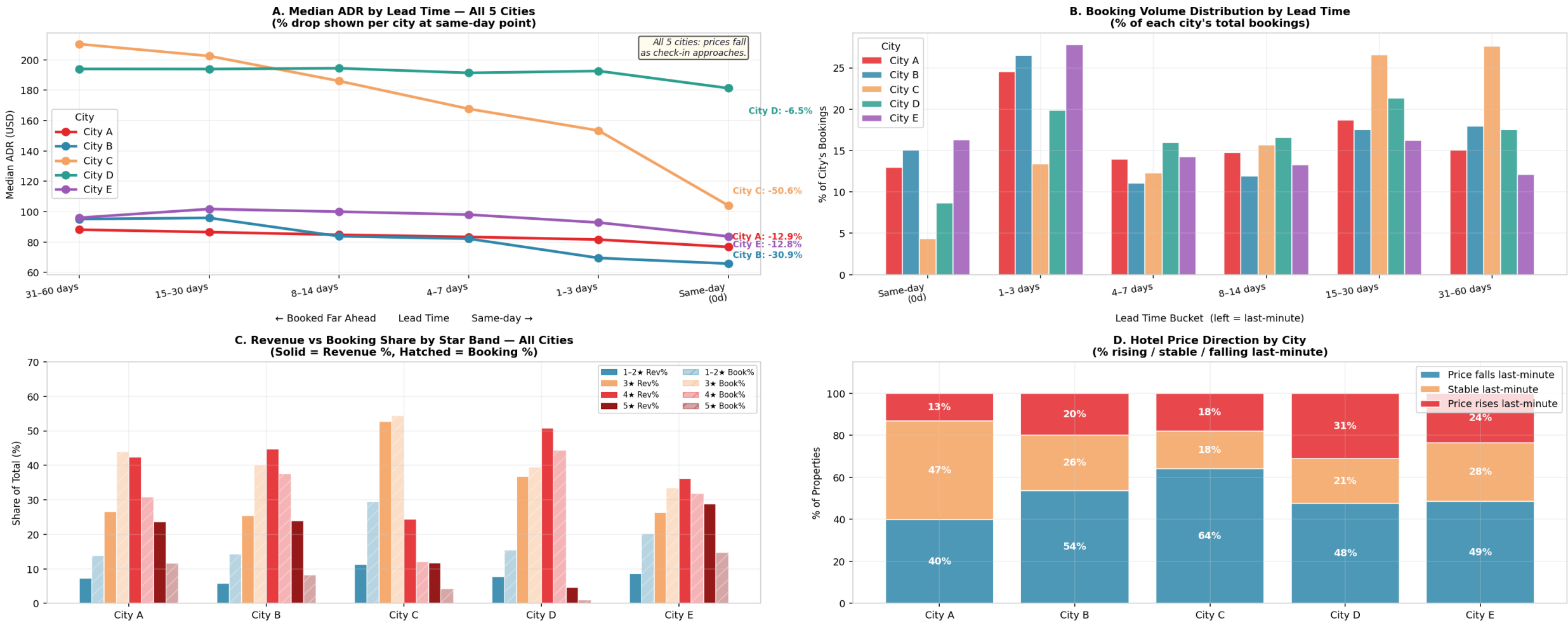
Cross-City Exploratory Data Analysis — All 5 Cities



- FigA: Two distinct market clusters: Budget/last-minute (A, B, E — median ADR \$79–\$94, High-Urgency 38–44%) vs Premium (C, D — median ADR \$190–\$192, High-Urgency 18–28%).
- FigD: Hotels are highest in demand across all cities except City B. In City B, Resorts are higher in demand.

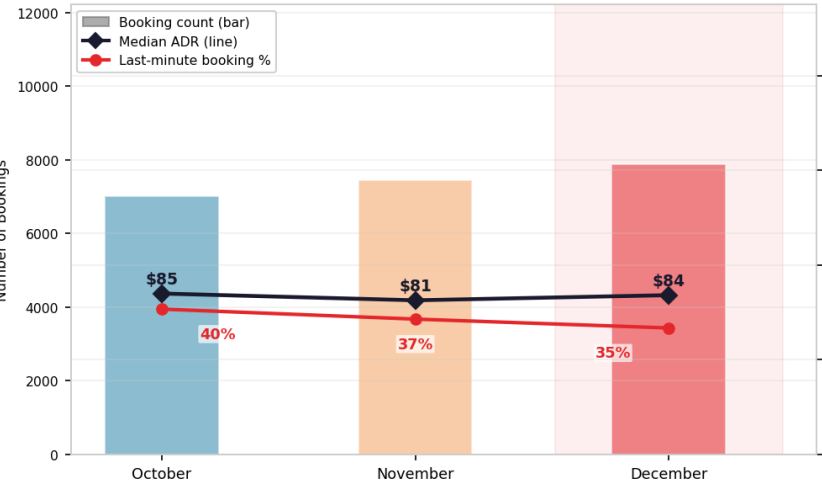
Cross-city analysis on slide 4 & 5— each slide shows one graph type across all 5 cities simultaneously.

Cross-City: Price Trends, Urgency Profile, Revenue by Segment & Price Direction

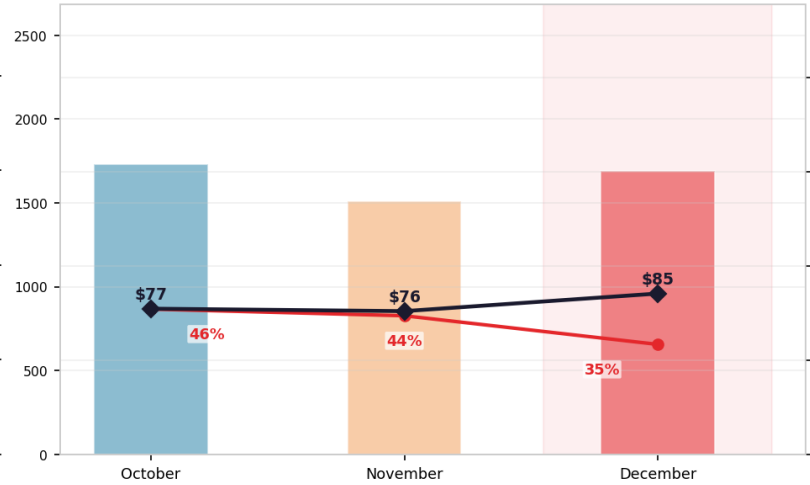


Monthly Booking Patterns — Oct, Nov & Dec Across All 5 Cities

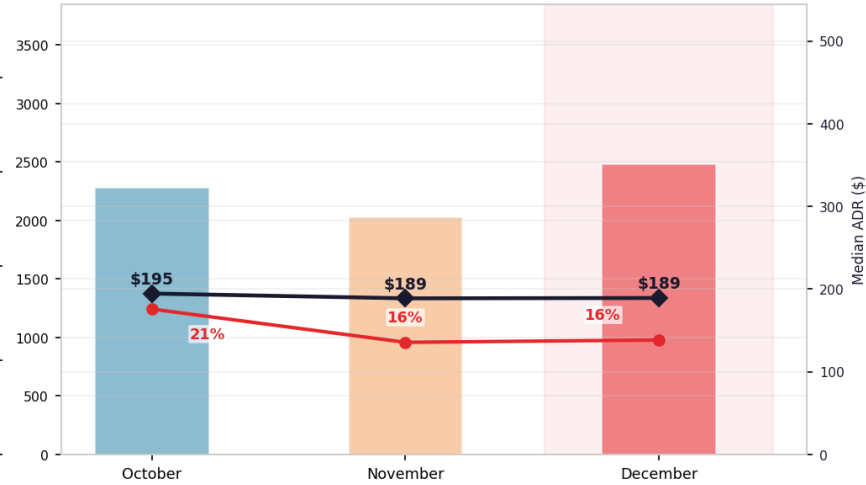
City A — Urban Mixed



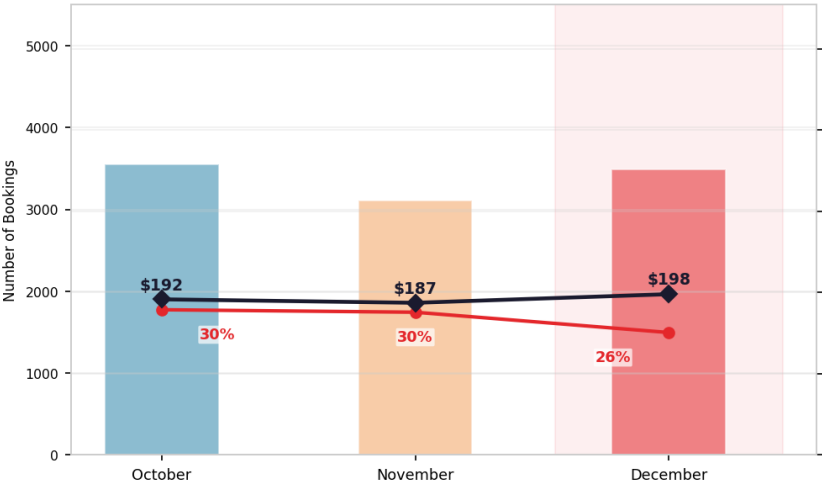
City B — Resort-Heavy



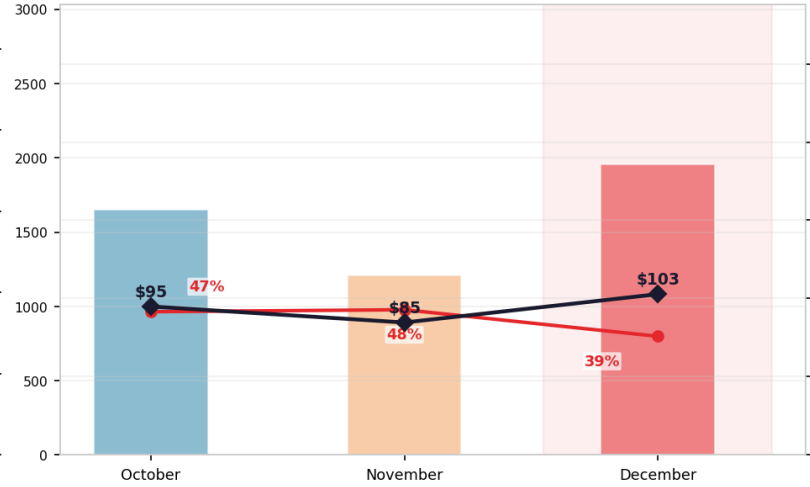
City C — Premium/Planned



City D — Premium/Stable



City E — Last-Minute



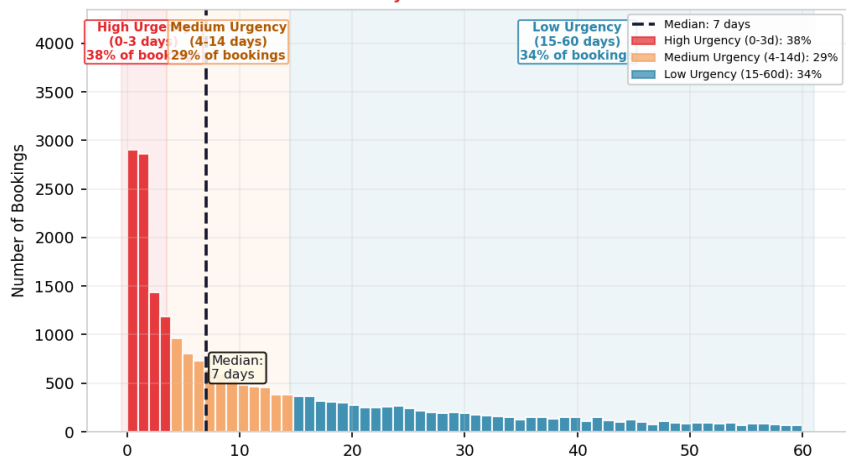
Monthly Booking Patterns

- * December ADR rises in leisure markets:
City B (resort): +10% vs October
City E (mixed resort): +8% vs October
City D (premium stable): +3% vs October
- * Last-minute bookings DROP in December:
People plan ahead for the holiday season.
Biggest drops: City B -11 percentage points,
City E -8 percentage points vs October.
- * December urgency strategy:
'Rooms filling up for the holidays'
or 'Prices rising - book now'
(scarcity framing works best in December)
- * October & November urgency strategy:
'Earn more loyalty points' works best -
last-minute rates are highest in these
months so booking is more impulsive

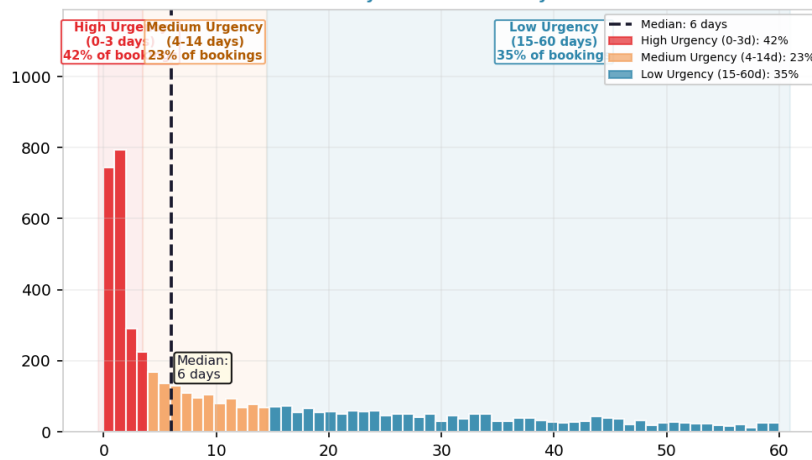
- December ADR rises in leisure markets (City B: +10%, City E: +8%, City D: +3%).
- Last-minute booking rate FALLS in December across all cities — people plan further ahead for holidays.
- Dec urgency strategy: holiday season scarcity/"price rising" framing.
- Oct/Nov: earn-sooner framing works best (last-min share highest).

Lead Time Distribution (Urgency Tiers) — All 5 Cities

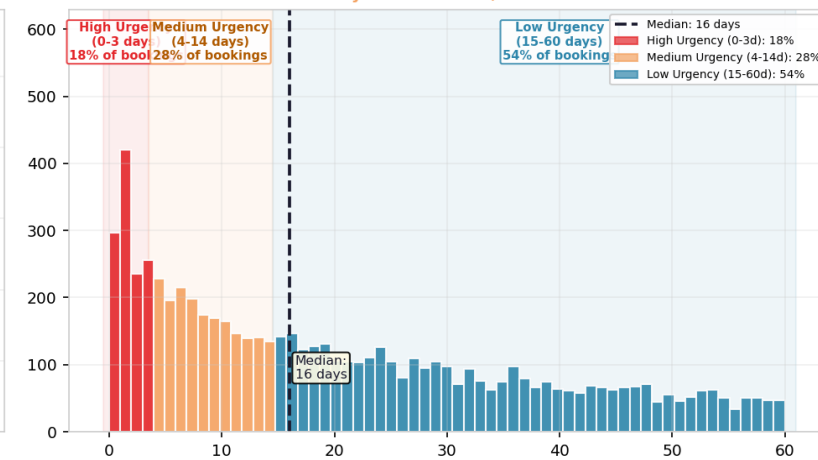
City A — Urban Mixed



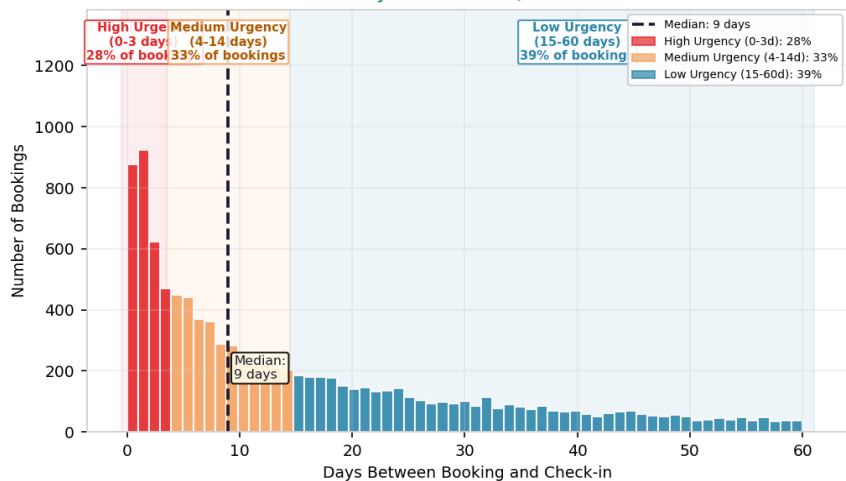
City B — Resort-Heavy



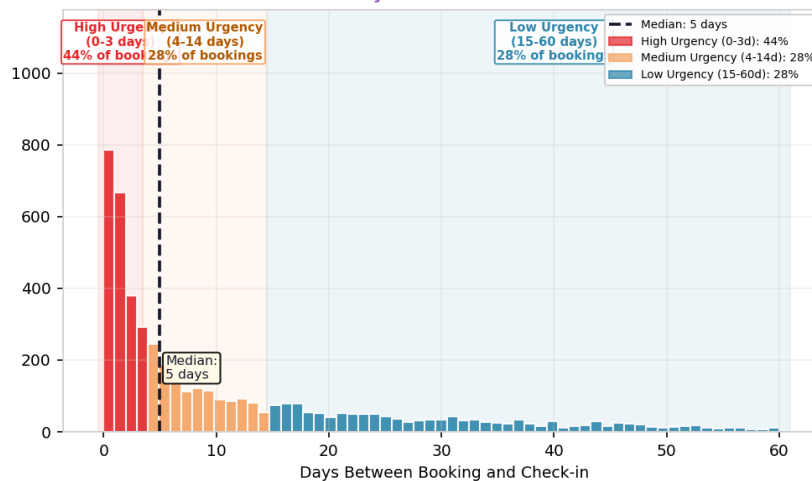
City C — Premium/Planned



City D — Premium/Stable



City E — Last-Minute

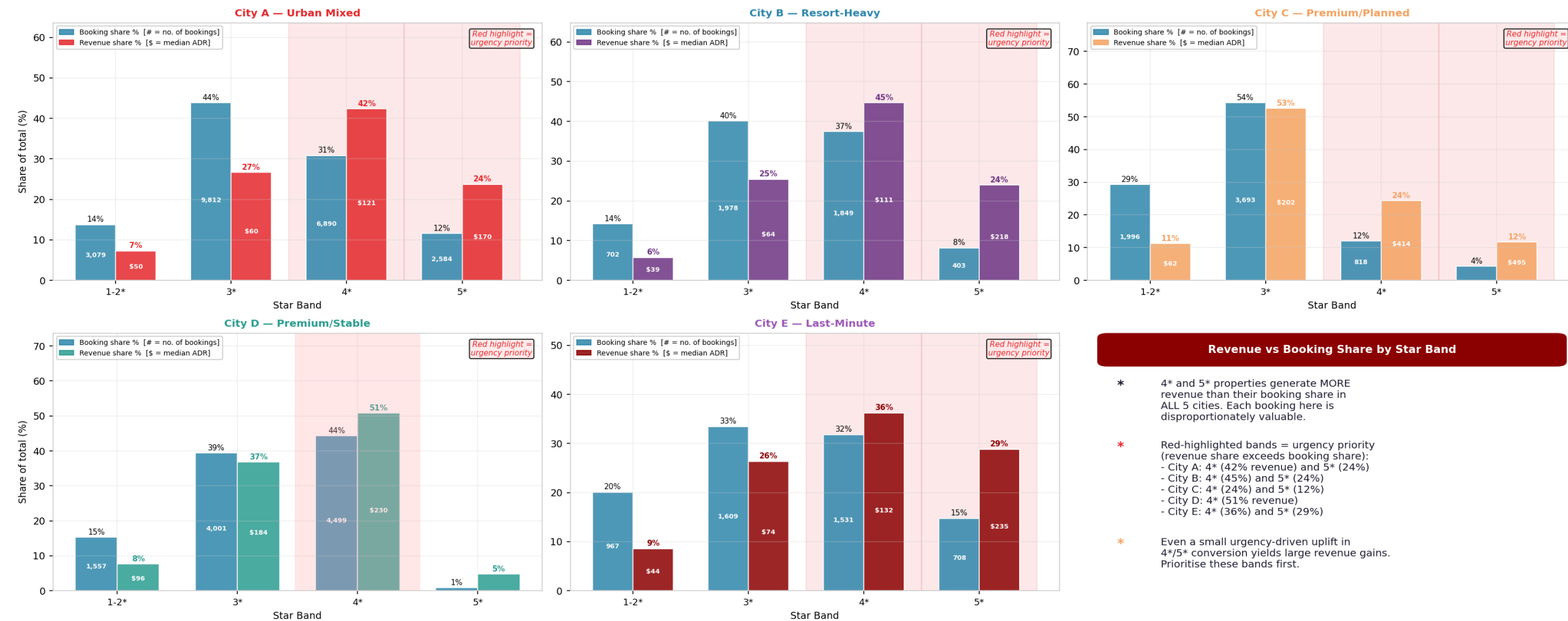


Lead Time and Urgency Tiers

- * City E has the MOST last-minute bookers: 44% of all bookings are made within 3 days of check-in. Needs the strongest time-triggered urgency messaging.
- * City B is close behind at 42% High-Urgency. Resort bookings cluster heavily around same-day and 1-3 days ahead.
- * City C is the most planned market: Only 18% High-Urgency - 54% of bookings made 15+ days ahead. Focus on medium and low urgency campaigns here.
- * Urgency message depends on tier:
High - Book tonight - prices lowest now
Medium - Lock in your rate before prices rise
Low - Secure your preferred room early

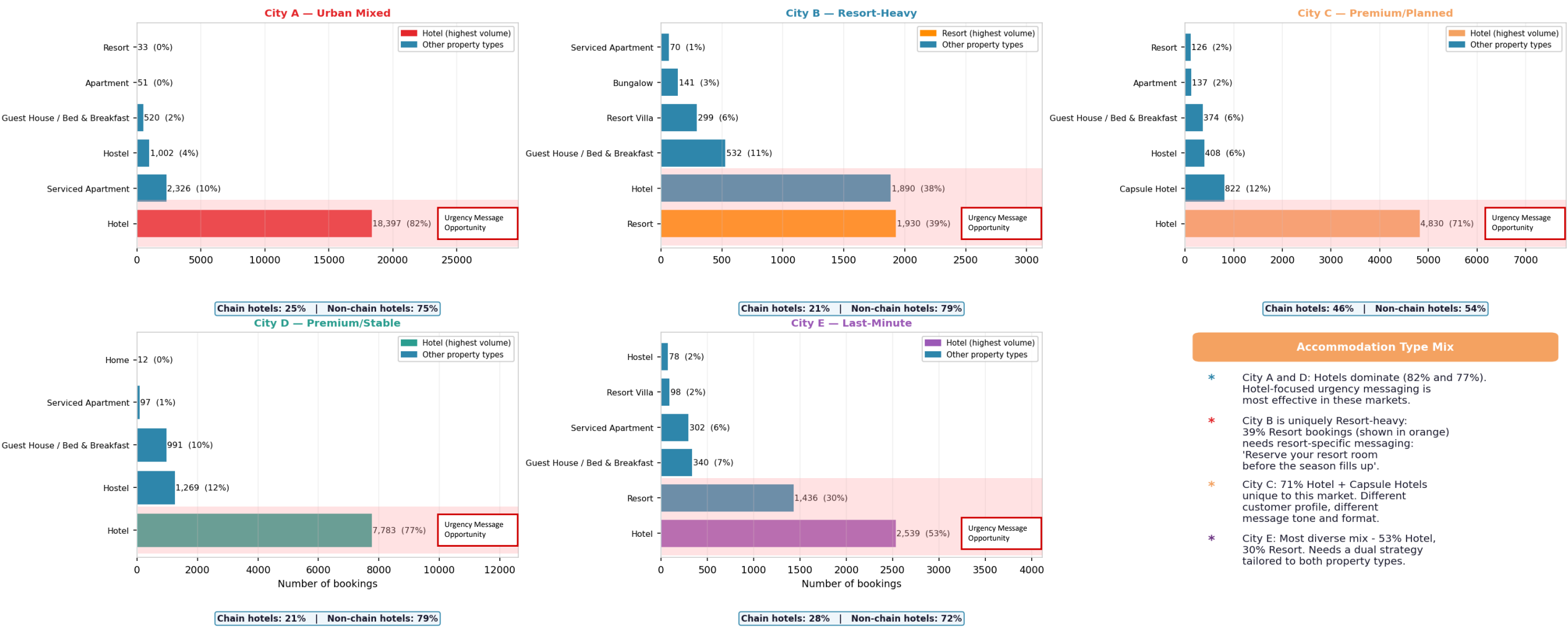
- City E (44% High) and City B (42% High) has potential for time-triggered messaging most urgently.
- City C (54% Low) needs longer lead-time campaigns.

Revenue vs Booking Share by Star Band — All 5 Cities



- Highlighted bands = revenue share > booking share (ratio >1x).
- 4★ is the priority urgency messaging segment in all cities. # inside blue bars = number of bookings; \$ inside revenue bars = median ADR.

Accommodation Type Mix — All 5 Cities



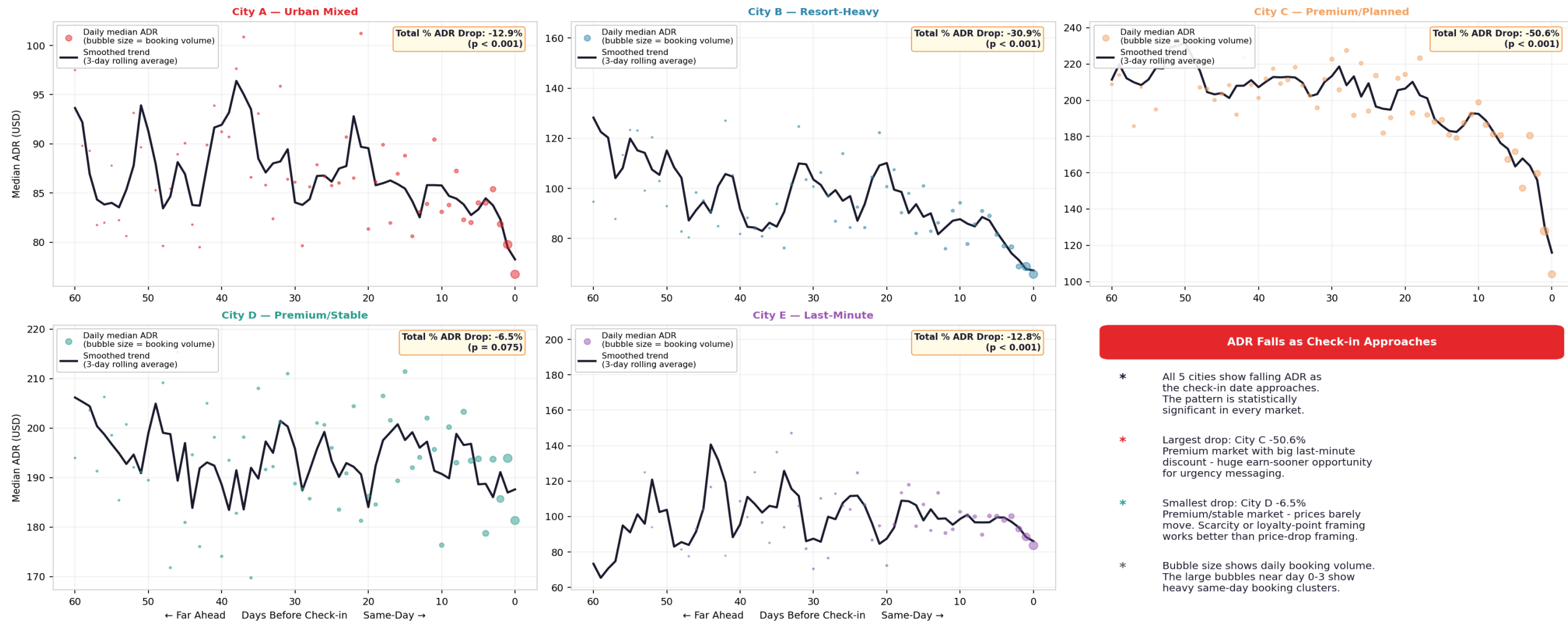
Accommodation Type Mix

- * City A and D: Hotels dominate (82% and 77%). Hotel-focused urgency messaging is most effective in these markets.
- * City B is uniquely Resort-heavy: 39% Resort bookings (shown in orange) needs resort-specific messaging: 'Reserve your resort room before the season fills up'.
- * City C: 71% Hotel + Capsule Hotels unique to this market. Different customer profile, different message tone and format.
- * City E: Most diverse mix - 53% Hotel, 30% Resort. Needs a dual strategy tailored to both property types.

- City-colour bar = highest-volume type (priority for urgency messaging reach).

- City B uniquely Resort-heavy (39%). City C has Capsule Hotels (unique market). City E has mixed Hotel+Resort split requiring dual strategy.

Median ADR by Days to Check-in — All 5 Cities

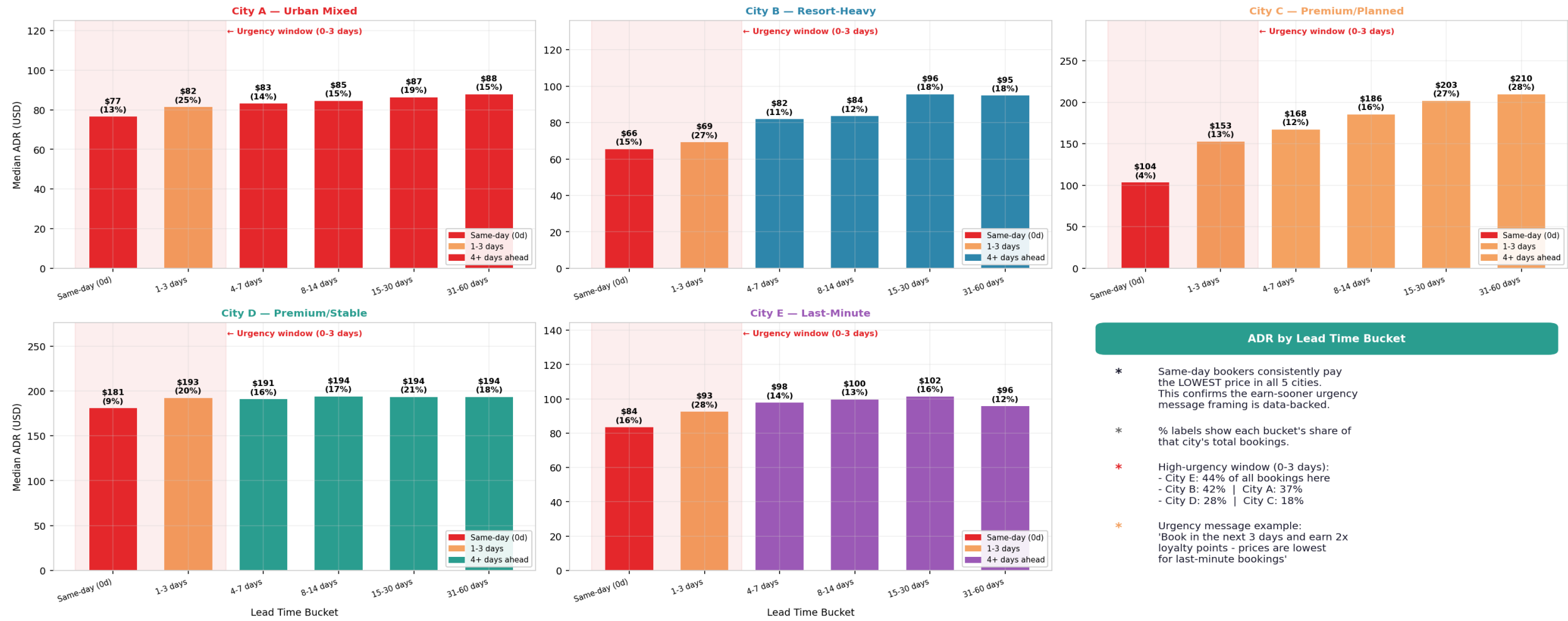


ADR Falls as Check-in Approaches

- * All 5 cities show falling ADR as the check-in date approaches. The pattern is statistically significant in every market.
- * Largest drop: City C -50.6% Premium market with big last-minute discount - huge earn-sooner opportunity for urgency messaging.
- * Smallest drop: City D -6.5% Premium/stable market - prices barely move. Scarcity or loyalty-point framing works better than price-drop framing.
- * Bubble size shows daily booking volume. The large bubbles near day 0-3 show heavy same-day booking clusters.

- Smoothed trend line shows price fall in all 5 cities as check-in approaches.
- % shown = ADR change from 31–60d to same-day.
- All results statistically significant ($p < 0.001$) except City D ($p = 0.075$ — most price-stable market).

Median ADR by Lead Time Bucket — All 5 Cities

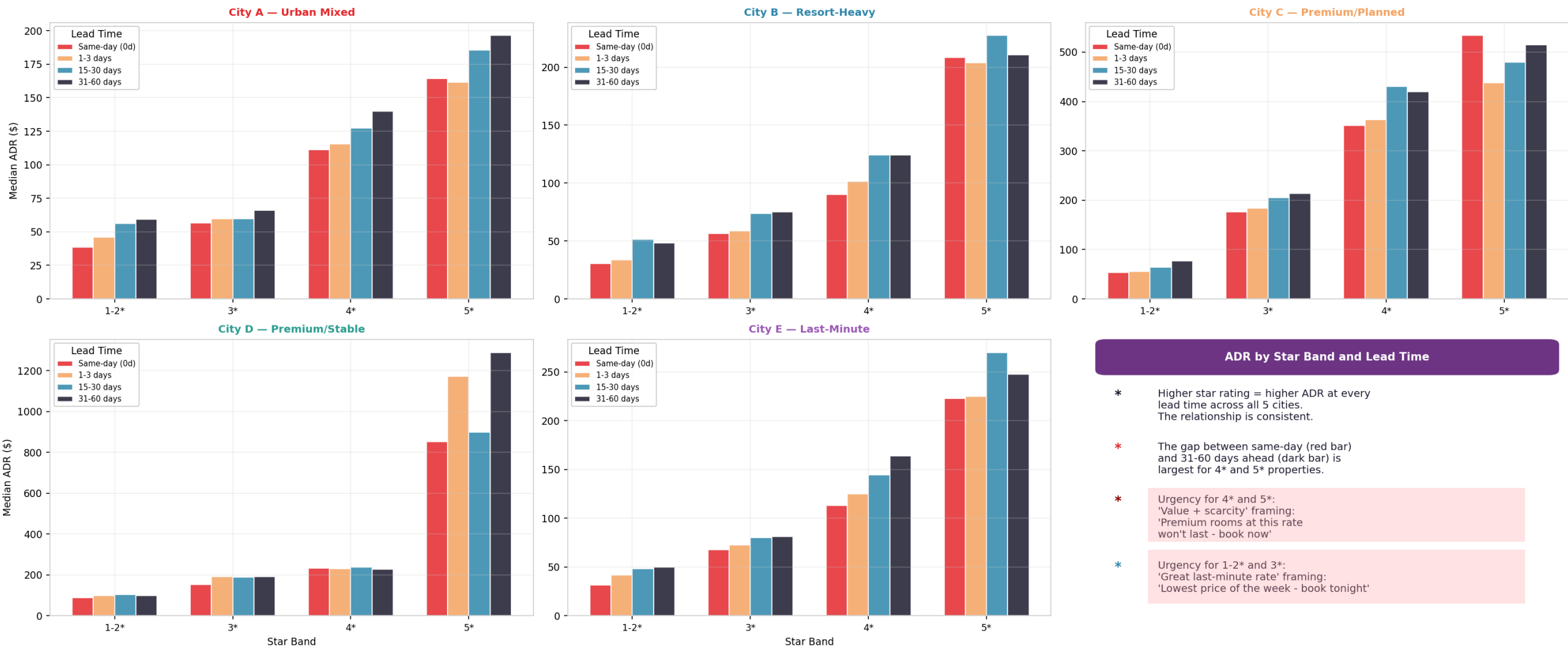


- Red = same-day (0d) · Amber = 1–3 days · City colour = 4+ days.

- % labels = share of city's total bookings.

- Same-day bookers consistently pay lowest ADR — validating earn-sooner or book-sooner urgency message framing.

ADR by Star Band and Lead Time — All 5 Cities



ADR by Star Band and Lead Time

- * Higher star rating = higher ADR at every lead time across all 5 cities. The relationship is consistent.
- * The gap between same-day (red bar) and 31-60 days ahead (dark bar) is largest for 4* and 5* properties.
- * Urgency for 4* and 5*: 'Value + scarcity' framing: 'Premium rooms at this rate won't last - book now'
- * Urgency for 1-2* and 3*: 'Great last-minute rate' framing: 'Lowest price of the week - book tonight'

- 4 lead time windows shown per star band.
- Higher star = higher ADR at every lead time in all cities.
- Gap between same-day and 31-60d is largest for 4★/5★ — these are the highest-value urgency targets.

ADR Trend by Star Band — All 5 Cities



ADR Trends by Star Band

- * % labels = total ADR drop from 31-60 days ahead to same-day for each star band.
- * Green shaded area = UPSELL WINDOW: where two adjacent star bands have similar prices - customers can be nudged to book a higher tier.
- * City A: 1-2* ≈ 3* at 15-60 days ahead - upsell budget bookers to mid-range when booking early.
- * City C: 4* ≈ 5* at 8-30 days - upgrade messaging works for premium customers booking in advance.
- * City D: 3* ≈ 4* across all lead times - a consistent upsell opportunity throughout the year.

Upsell windows:

- City C: 4★≈5★ at 8–30d
- City D: 3★≈4★ across (almost) all lead times
- City A & E: 1–2★≈3★ at 15-60d & 4–14d, respectively.

Urgency Messaging Strategy by Segment

Segments by price behaviour × revenue concentration × booking timing. Use price-rise claims only where the property-level data supports them; otherwise use value or availability framing.

Strategy segment	Target cities, star bands & properties	Analytical signal	Message to test	Funnel & success metric
Dynamic price-rise only	Rising-property cohort across all cities; prioritize City D/E/B and 4★–5★ listings.	Only 109 / 565 comparable properties rise >5% last-minute. Rising share: City D 31%, City E 24%, City B 20%.	“Rates for this property have increased recently — book now to secure today’s price.”	Property page + booking form. KPI: conversion uplift, ADR / revenue per visitor
Premium revenue protectors	4★/5★ in Cities A, B, D, E; City C’s 3★ segment; chain and premium hotels.	Premium hotels drive outsized value: 4★/5★ properties are 40.1% of bookings but 56.5% of booking value. Prioritize 4★ messaging in Cities A/B/D/E; in City C, prioritize 3★ because it contributes 52.7% of booking value.	“High-demand room at the current rate — secure it before availability changes.”	Search results + property page. KPI: revenue per visitor and premium booking conversion.
Last-minute value markets	Cities E, B, A; high-volume in Hotels / Resorts / Serviced Apartments; 3★–4★ where reach is highest.	Bookings within 3 days: City E 44%, City B 42%, City A 38%. ADR still falls: E –13%, B –31%, A –13%.	“Great last-minute value available — lock in this deal today.” Avoid saying prices are rising.	Search results page. KPI: click-through to property page, conversion, cancellation
Stable / volatile premium	City D, especially 4★ properties; stable or rising properties in all cities; chain hotels.	City D has smallest ADR drop (–6.5%) and highest rising-property share (31%); City D 4★ is slightly up (+1.6%).	“Popular property — availability can change quickly.” Use price-rise wording only if property flag is active.	Property page + booking form. KPI: booking completion rate and abandonment reduction.
Budget & falling-price long tail	1–2★ and many 3★ properties; non-chain long-tail hotels; properties classified as falling or stable.	Budget and non-chain segments see large same-day ADR drops: 1-2★ - 35%, 3★ -54.3%, non-chain -35.8%. Most properties, 80.7%, are flat or falling last-minute	Use value framing: “Good value for this area” / “Flexible option”; avoid pressure-based price-rise claims.	Search results filters / sort modules. KPI: engagement and conversion without trust deterioration.

Implementation principle: build a property-level price-direction flag refreshed daily. If confidence is high and recent price movement is rising, use price-protection language; otherwise default to availability/value language and validate via city-level A/B tests.

Source: 49,061 valid bookings across 5 cities; 3 invalid negative-lead-time records excluded. Price-direction classification compares median last-minute ADR (0–3d) vs early ADR (15+d) with ±5% threshold.

Recommended Next Steps

1

A/B Test Urgency Message Types

Run controlled tests per segment and city — e.g: compare conversion with earn-sooner vs availability vs value-framing messages.

2

Expand to Full Year Dataset

Seasonal effects (peak vs off-peak) may significantly change price dynamics and urgency effectiveness.

3

Build Real-Time Price Signal

Develop a live model flagging which properties have rising vs falling prices — enabling dynamic, truthful urgency messages per listing.

4

City-Level Calibration

City C's -50.6% drop vs City D's -6.5% requires very different urgency intensity per market. Customize urgency messaging before rolling out.

What We Found & What It Means

All 5

Cities confirm prices
fall last-minute

34%

Bookings made within
3 days of check-in

–50.6%

Biggest market drop
(City C, premium segment)

Urgency messaging can improve conversion across all 5 markets — but only when the message is matched to the actual price signal and calibrated per city.